NSIS Policy Projection

Mathematical Functions

Employment injury, maternity, unemployment and old-age policy scenarios

Model basis	Annual deterministic projection
Time index	t = 0, 1, ..., T - 1
Policy index	j in {E, M, U, P}

Reference format: functions and component definitions only

Notation Conventions

Symbol	Meaning
t	Projection-year index, beginning at 0.
T	Number of projected years.
j	Policy branch: E = employment injury, M = maternity, U = unemployment, P = old-age pension.
Rates	Percentages are represented as decimals; for example, 2% = 0.02.
Tk	Bangladeshi Taka.

Core Projection Functions

1. Covered-worker projection

$$N_t = N_0(1 + g_N)^t$$

Component	Definition
N_t	Covered workers in year t.
N_0	Covered workers in the start year.
g_N	Annual growth rate of covered workers.
t	Projection-year index.

2. Average monthly wage projection

$$W_t = W_0(1 + g_W)^t$$

Component	Definition
W_t	Average monthly wage in year t.
W_0	Average monthly wage in the start year.
g_W	Annual wage-growth rate.
t	Projection-year index.

3. Annual covered payroll and contribution base

$$X_t = P_t = 12 N_t W_t$$

Component	Definition
X_t	Contribution base used by each policy branch in year t.
P_t	Annual covered payroll in year t.
12	Months in a year.
N_t	Covered workers in year t.
W_t	Average monthly wage in year t.

Expected Claim Functions

4. Employment injury claims

$$Q_t^E = N_t q^E$$

Component	Definition
Q_t^E	Expected employment injury claims in year t.
N_t	Covered workers in year t.
q^E	Annual employment injury claim rate per covered worker.

5. Maternity claims

$$Q_t^M = N_t s^F b$$

Component	Definition
Q_t^M	Expected maternity claims in year t.
N_t	Covered workers in year t.
s^F	Female share of covered workers.
b	Annual birth rate among covered female workers.

6. Unemployment claims

$$Q_t^U = N_t u$$

Component	Definition
Q_t^U	Expected unemployment claims in year t.
N_t	Covered workers in year t.
u	Annual unemployment claim rate per covered worker.

7. Old-age beneficiaries

$$Q_t^P = N_t r$$

Component	Definition
Q_t^P	Expected old-age pension beneficiaries in year t.
N_t	Covered workers in year t.
r	Retiree ratio applied to covered workers.

Benefit-Cost Functions

8. Employment injury base cost

$$B_t^E = Q_t^E W_t m^E$$

Component	Definition
B_t^E	Employment injury benefit cost before administrative margin.
Q_t^E	Expected employment injury claims.
W_t	Average monthly wage.
m^E	Average benefit cost expressed as wage months per claim.

9. Maternity base cost

$$B_t^M = Q_t^M W_t \left(\frac{L}{4.345}\right) \rho^M$$

Component	Definition
B_t^M	Maternity benefit cost before administrative margin.
Q_t^M	Expected maternity claims.
W_t	Average monthly wage.
L	Paid maternity-leave duration in weeks.
4.345	Average weeks per month used to convert leave weeks to wage months.
rho^M	Maternity wage-replacement rate.

10. Unemployment base cost

$$B_t^U = Q_t^U \left(W_t m^U \rho^U + c^U\right)$$

Component	Definition
B_t^U	Unemployment benefit and service cost before administrative margin.
Q_t^U	Expected unemployment claims.
W_t	Average monthly wage.
m^U	Number of monthly cash-benefit payments.
rho^U	Unemployment wage-replacement rate.
c^U	Fixed employment-service cost per claimant.

11. Old-age pension base cost

$$B_t^P = Q_t^P W_t (12) \rho^P$$

Component	Definition
B_t^P	Annual old-age pension cost before administrative margin.
Q_t^P	Expected old-age pension beneficiaries.
W_t	Average monthly wage.
12	Monthly payments in a year.
rho^P	Pension replacement rate relative to average wage.

12. Final annual policy cost

$$C_t^j = B_t^j(1 + a)$$

Component	Definition
C_t^j	Final annual cost for policy branch j in year t.
B_t^j	Base benefit cost for policy branch j in year t.
a	Administrative, contingency and reserve margin.

Financing and Reserve Functions

13. Annual contribution income

$$K_t^j = X_t \tau^j$$

Component	Definition
K_t^j	Contribution income for policy branch j in year t.
X_t	Total covered payroll contribution base.
tau^j	User-set contribution or premium rate for policy branch j.

14. Accumulated reserve balance

$$R_t^j = R_{t-1}^j(1 + i) + K_t^j - C_t^j, \qquad R_{-1}^j = 0$$

Component	Definition
R_t^j	End-of-year accumulated reserve for policy branch j.
R_{t-1}^j	Previous year-end reserve.
i	Annual fund investment-return rate.
K_t^j	Contribution income in year t.
C_t^j	Final annual cost in year t.
R_{-1}^j	Initial reserve before the first projection year, set to zero.

15. Cumulative cost

$$CC_t^j = \sum_{k=0}^{t} C_k^j$$

Component	Definition
CC_t^j	Cumulative policy cost from the start year through year t.
C_k^j	Final annual cost in projection year k.
k	Summation index.

16. Cumulative contributions

$$CK_t^j = \sum_{k=0}^{t} K_k^j$$

Component	Definition
CK_t^j	Cumulative contribution income from the start year through year t.
K_k^j	Contribution income in projection year k.
k	Summation index.

Required Premium-Rate Functions

17. Final-year accumulation weight

$$d_t = (1 + i)^{T - 1 - t}$$

Component	Definition
d_t	Factor that accumulates year-t cash flow to the end of the projection.
i	Annual fund investment-return rate.
T	Number of projection years.
t	Projection-year index.

18. Raw required premium or contribution rate

$$\widehat{\tau}^{j} = \frac{\sum_{t=0}^{T-1} C_t^j d_t}{\sum_{t=0}^{T-1} X_t d_t}$$

Component	Definition
widehat{tau}^{,j}	Unrounded rate that finances projected costs by the end of the projection.
C_t^j	Final annual cost for policy branch j.
X_t	Total covered payroll contribution base.
d_t	Final-year accumulation weight.
T	Number of projection years.

19. Displayed required rate

$$\tau_{\mathrm{req}}^{j} = \frac{\lceil 10{,}000\widehat{\tau}^{j} \rceil}{10{,}000}$$

Component	Definition
tau_{mathrm{req}}^j	Required premium rate displayed by the calculator.
widehat{tau}^{,j}	Raw required rate.
lceilcdotrceil	Ceiling function: rounds upward to the next integer.
10{,}000	Precision factor producing upward rounding to four decimal places in rate form.

A displayed rate of 0.0123 is shown to the user as 1.23%.

20. Final-year payroll output

$$P^F = 12N_0(1+g_N)^{T-1}W_0(1+g_W)^{T-1}$$

Component	Definition
P^F	Annual covered payroll in the final projection year.
N_0	Start-year covered workers.
g_N	Annual worker-growth rate.
W_0	Start-year average monthly wage.
g_W	Annual wage-growth rate.
T	Number of projection years.

Policy-Index Mapping

Index	Policy branch	Primary claim/cost inputs
E	Employment injury	q^E, m^E
M	Maternity	s^F, b, L, rho^M
U	Unemployment	u, m^U, rho^U, c^U
P	Old-age pension	r, rho^P